

AI Text Detector: Distinguishing Human vs. AI-Generated Text with Stylometric Features

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Abstract

Large language models (LLMs) such as ChatGPT have made it trivial to generate fluent English text at scale. This creates new opportunities, but also raises concerns for academic integrity, misinformation, and the authenticity of online content. In this project, we investigate whether it is possible to distinguish human-written texts from AI-generated texts using only interpretable stylometric features and traditional machine learning models, rather than another large black-box neural network. We construct a mixed corpus of roughly 2,500 human and 2,500 AI texts by combining real essays and question-answer pairs with data derived from the HC3 human-ChatGPT comparison corpus and additional synthetic examples. On top of this corpus we design a compact set of linguistic “fingerprints” that capture vocabulary richness, sentence rhythm, punctuation patterns, readability, and the use of typical “AI buzzwords”, inspired by classical stylometry and authorship attribution research. We then train a soft-voting ensemble of five classifiers (Random Forest, Gradient Boosting, Logistic Regression, RBF-SVM, and a shallow MLP) and evaluate it on a held-out test set of 998 samples. Our system achieves approximately 91% accuracy, 92.6% precision and 89.4% recall for the AI class, with ROC AUC of 0.968 and PR AUC of 0.971. We further analyze feature importance and case studies, showing that structural aspects such as readability, average word length, and punctuation density are particularly informative. Overall, the results suggest that a transparent stylometric approach can provide competitive detection performance while remaining lightweight and interpretable.

1 Introduction

Large language models (LLMs) such as ChatGPT, GPT-4, and their successors can now generate fluent English text on almost any topic with very little effort from the user. This capability has rapidly diffused into everyday applications, from writing assistance and summarization to code generation and content creation. While these systems are useful, they also raise new risks: students can generate essays or lab reports without doing the work, malicious actors can mass-produce synthetic news articles, and it becomes increasingly difficult to tell whether a document was authored by a human or a model.

These concerns have led to an emerging research area on AI-generated text detection, which seeks to distinguish machine-generated content from human writing [4, 5]. At the same time, work in stylometry and authorship attribution has a long history of using quantitative features to characterize individual writing styles [1, 2]. Our project lies at the intersection of these two lines of work.

The central research question we address is:

Can we reliably distinguish human-written texts from AI-generated texts using only interpretable stylometric features and classical machine learning models?

Rather than relying on a large transformer-based detector, we deliberately design a small set of hand-crafted features that capture the “shape” of the text: vocabulary richness, sentence length variation, punctuation patterns, readability, and the frequency of typical AI-style terms. These features form a compact

representation that is easier to understand and communicate to non-expert stakeholders such as instructors and students.

Our main contributions are:

- We curate a mixed corpus of human and AI texts by combining real essays and responses with portions of the HC3 human–ChatGPT comparison dataset [8], plus additional synthetic examples to broaden topic coverage.
- We design a stylometric feature set that summarizes vocabulary usage, sentence rhythm, discourse markers, readability, and the use of characteristic “AI buzzwords”, inspired by classical authorship attribution features.
- We train an ensemble of five traditional classifiers and show that, on a held-out set of 998 texts, the detector reaches 91.1% accuracy, ROC AUC of 0.968, and PR AUC of 0.971, while remaining fast enough for interactive use.
- We analyze feature importance and several case studies to understand why the system makes particular decisions, highlighting how style and structure differ between human and AI-generated text.

The rest of the report is organized as follows. Section 2 reviews related work on AI text detection and stylometry. Section 3 formalizes the task and describes our system design. Section 4 presents the dataset, experimental setup, and quantitative results. Section ?? discusses strengths, limitations, and possible extensions, and Section 6 concludes.

2 Related Work

2.1 AI-Generated Text Detection

Early work on detecting neural text focused on language model perplexity and simple statistical signatures. Gehrmann et al. introduced GLTR, a visual tool that highlights tokens based on their likelihood under a language model and shows that human-written text tends to contain more low-probability words than model-generated text, enabling simple detection heuristics [4]. More recently, Mitchell et al. proposed DetectGPT, which uses the curvature of a model’s log-probability surface under perturbations to discriminate machine-generated from human text in a zero-shot setting [5]. These methods typically assume access to the generating model or a similar large language model for scoring.

Another line of work explores watermarking LLM outputs [6], where the model is modified to prefer tokens from a secret green list so that generated texts carry an embedded statistical signal. Watermarks can achieve high detection accuracy when the text is not heavily edited, but their robustness under paraphrasing and human rewriting is still an active research topic [7].

More traditional supervised approaches treat detection as a binary classification problem, training transformers such as RoBERTa on datasets containing both human and AI texts. For example, Guo et al. introduced the HC3 corpus of human and ChatGPT answers across multiple domains and evaluated both GLTR-style logistic regression and fine-tuned transformers as detectors [8]. Follow-up work has extended HC3 to additional tasks and settings [9].

Our work differs from these approaches in two main aspects. First, we intentionally restrict ourselves to a small, interpretable feature set computed from the raw text, rather than using hidden states of a large transformer. Second, we focus on classical machine learning models that can be trained and deployed on modest hardware, while still achieving competitive performance on a realistic mixed dataset.

2.2 Stylometry and Authorship Attribution

Stylometry studies an author’s writing style using quantitative features such as word frequencies, sentence statistics, and syntactic patterns. These features have been widely used for authorship attribution, verifica-

Figure 1: Conceptual overview of the AI text detection pipeline: collection, preprocessing, feature extraction, ensemble training, and prediction. (Placeholder figure.)

tion, and profiling [1, 2]. Common stylometric features include lexical statistics (e.g., vocabulary richness, function word frequencies), structural features (e.g., sentence and paragraph lengths), and sometimes higher-level syntactic and semantic features [3].

Stamatatos provides a comprehensive survey of modern authorship attribution methods, emphasizing that relatively simple features like the frequencies of common words and character n -grams often capture strong stylistic signals [1]. More recent surveys by He et al. and others review the use of stylometric features in various author-related tasks and discuss the transition towards neural representations while still highlighting the value of interpretable features [2].

Our feature design is directly inspired by this classical stylometry literature. Instead of aiming to identify a specific author, we use stylometric features to separate two broad “authors”: humans and large language models. This allows us to leverage decades of work on authorship analysis in a new context.

3 Problem Definition and Methodology

3.1 Task Definition

We formulate AI text detection as a binary classification problem. Given an input text x (typically a paragraph or short essay), the goal is to predict a label $y \in \{0, 1\}$, where $y = 0$ denotes human-written text and $y = 1$ denotes AI-generated text.

Instead of operating directly on token sequences, our system first maps x to a fixed-dimensional feature vector $\phi(x) \in \mathbb{R}^d$ capturing stylometric properties. A classifier f then outputs an estimated probability

$$\hat{p}(y = 1 \mid x) = f(\phi(x)) \in [0, 1],$$

which we interpret as the AI probability. A final decision is obtained by thresholding $\hat{p}$ (e.g., at 0.5), but in many applications the probability itself is informative.

3.2 System Overview

1. **Data collection and labeling:** We aggregate texts from multiple sources and label them as human or AI based on origin (e.g., human-written essays vs. ChatGPT responses).
2. **Preprocessing:** We clean the texts, remove extremely short or noisy samples, and normalize whitespace.
3. **Feature engineering:** For each text, we compute a vector of stylometric features that constitute its linguistic fingerprint.
4. **Model training:** We train an ensemble of five classifiers on the feature vectors using stratified cross-validation.
5. **Evaluation and analysis:** We evaluate on a held-out test set, compute standard metrics and AUC scores, and analyze feature importances and example predictions.
6. **Interactive prediction:** A small interface accepts arbitrary text, runs the same feature extractor, and returns an AI probability along with an explanation based on feature values.

3.3 Stylometric Feature Engineering

Our feature extractor computes a compact set of interpretable statistics for each input text. We group them into several categories.

Lexical richness. The *Unique Word Ratio* measures vocabulary diversity as the number of distinct tokens divided by the total token count. Human authors often use more idiosyncratic and personal wording, while AI systems tend to reuse safe, generic phrases. We also compute a simple *Lexical Diversity* index based on type–token ratio scaled by text length.

AI-style vocabulary. We define an *AI Word Ratio* that counts how many tokens fall into a manually curated list of “AI-sounding” terms commonly observed in LLM outputs, such as *paradigm*, *framework*, *moreover*, *furthermore*, and *in addition*. While this list is not exhaustive, it captures a recognizable register of formal, templated academic language.

Sentence rhythm and structure. We approximate *Sentence Length Variation* as the standard deviation of sentence lengths in tokens, normalized by the mean. Human texts typically mix short and long sentences, whereas AI outputs often exhibit more regular, medium-length sentences. We also compute *Average Word Length* as a proxy for formality and *Burstiness*, which measures how much information is concentrated in a few long sentences versus spread across the text.

Punctuation and discourse markers. *Punctuation Density* is defined as the proportion of punctuation characters in the text. Human writing often exhibits irregular use of commas, dashes, and exclamation marks, while AI tends to produce more standardized punctuation. We also compute a *Conjunction / Transition Ratio*, measuring how frequently connectors such as *however*, *moreover*, and *in addition* occur relative to the text length.

Readability and style. We implement a simplified *Readability Score* inspired by Flesch–Kincaid-style formulas, combining average sentence length and average syllables per word into a normalized index. In addition, we track the *Contraction Ratio* (frequency of forms like *don’t*, *I’m*) and a rough *Passive Voice Ratio* using heuristic pattern matching. Informal human writing tends to contain more contractions and varied voice, whereas AI essays are often consistently formal.

These features are deliberately simple and transparent. When the detector flags a text as AI-generated, we can often explain the decision in terms of a few salient feature values, such as unusually long average word length, high AI word ratio, or very low sentence length variation.

3.4 Model Architecture

We use an ensemble of five base classifiers, each trained on the same feature vectors:

- **Random Forest (RF):** An ensemble of decision trees trained with bootstrap samples and feature subsampling.
- **Gradient Boosting (GB):** A boosting model that sequentially fits trees to residual errors.
- **Logistic Regression (LR):** A linear classifier with ℓ_2 regularization, providing a simple baseline and well-calibrated probabilities.
- **Support Vector Machine (SVM)** with RBF kernel: A non-linear classifier operating in a kernel-induced feature space.
- **Multi-Layer Perceptron (MLP):** A small feed-forward neural network with one hidden layer.

Each base model outputs an estimated probability $p_k(y = 1 \mid \phi(x))$. The ensemble combines these by soft voting:

$$\hat{p}(y = 1 \mid x) = \sum_{k=1}^K w_k p_k(y = 1 \mid \phi(x)),$$

where $K = 5$ and $\sum_k w_k = 1$. In our implementation we use equal weights for simplicity, as we did not observe substantial gains from manual reweighting.

The Random Forest component additionally provides feature importance scores based on impurity reduction, which we use in Section 4.6 to analyze which features carry the most signal.

3.5 Implementation Details

We implement the system in Python using `scikit-learn`, `pandas`, `NumPy`, and `nltk` for tokenization and sentence splitting. Reproducibility is enforced via a global random seed shared across data splitting and model initialization. Data and model artifacts are stored under a simple directory structure (`data/`, `models/`, `figures/`), and the main script exposes several options: creating the dataset, training the model, evaluating on the test set, and running an interactive prediction mode.

4 Experiment Design and Evaluation

4.1 Dataset Construction

Our dataset combines both real and synthetic sources to capture a range of writing styles.

Real data. We collect human-written essays and short answer texts from course-related materials and publicly available sources with appropriate licenses. For AI texts, we use answers generated by ChatGPT for prompts similar to those seen in our courses, as well as samples from the HC3 dataset [8], which pairs human expert responses with ChatGPT responses across multiple domains. HC3 has been used in several prior detection studies and provides a realistic benchmark for human vs. ChatGPT discrimination [8, 9].

Synthetic data. To augment topic diversity and avoid overfitting to a narrow domain, we generate additional synthetic texts using simple templates. For human-like texts, templates mimic informal diary entries or opinionated comments. For AI-like texts, templates emulate formal academic language with generic structures, such as “The multifaceted nature of X necessitates a comprehensive reevaluation of Y”. While these synthetic samples are not as rich as real data, they help the model see a wider range of surface forms.

After cleaning and filtering out extremely short texts (less than 30 characters), we merge the real and synthetic sources into a single corpus. The final dataset contains 4,988 labeled texts: 2,489 human ($y = 0$) and 2,499 AI ($y = 1$). This near-perfect class balance simplifies evaluation and avoids trivial baseline issues.

4.2 Preprocessing

Preprocessing consists of the following steps:

- Normalize whitespace and remove non-printable characters.
- Discard samples with missing text or extremely short length.
- Optionally truncate very long texts at a fixed word limit to control feature ranges.
- Apply sentence tokenization and word tokenization for feature computation.

We do not perform stemming or lemmatization, as our features are primarily based on lengths, counts, and fixed word lists rather than lexical semantics.

4.3 Train–Test Split and Cross-Validation

We perform a stratified 80/20 split at the text level, resulting in 3,990 training samples and 998 test samples. The split preserves the approximate 50/50 ratio of human and AI texts, with 1,991 human and 1,999 AI examples in the training set, and 498 human and 500 AI examples in the test set.

On the training set, we run 5-fold stratified cross-validation to estimate generalization performance and tune a small number of hyperparameters (e.g., number of trees in the Random Forest, learning rate for Gradient Boosting, regularization strength for Logistic Regression). The final reported metrics in this section are calculated on the held-out test set.

4.4 Evaluation Metrics

We evaluate the detector using standard classification metrics:

- **Accuracy**: proportion of correctly classified samples.
- **Precision** (for the AI class): fraction of predicted AI texts that are actually AI.
- **Recall / Sensitivity** (for the AI class): fraction of actual AI texts that the model correctly detects.
- **F1-score**: harmonic mean of precision and recall.

To assess performance across different decision thresholds, we also compute:

- **ROC AUC** (Area Under the Receiver Operating Characteristic Curve): probability that a randomly chosen AI sample receives a higher AI score than a randomly chosen human sample.
- **PR AUC** (Area Under the Precision–Recall Curve) for the AI class: summarizes the trade-off between precision and recall as the threshold varies, which is especially informative when the positive class is of particular interest.

Finally, we inspect the confusion matrix to break down true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), and we analyze the distribution of prediction confidences.

4.5 Quantitative Results

On the held-out test set of 998 samples, the ensemble achieves the following performance:

- Accuracy: 91.08%
- Precision (AI): 92.55%
- Recall (AI): 89.40%
- F1-score (AI): 90.95%
- ROC AUC: 0.968
- PR AUC (AI): 0.971

The confusion matrix is:

	Predicted Human	Predicted AI
True Human	462	36
True AI	53	447

This corresponds to a false positive rate (human $\rightarrow$ AI) of approximately 7.2% and a false negative rate (AI $\rightarrow$ human) of about 10.6%. In other words, most human texts are correctly recognized as human, and most AI texts are correctly flagged as AI, with remaining errors concentrated in borderline cases where the style is ambiguous.

The ROC curve lies well above the diagonal baseline, and the ROC AUC of 0.968 indicates that in roughly 96.8% of random human–AI text pairs, the AI text receives a higher AI score than the human text. The PR AUC of 0.971 reflects that the detector maintains high precision even as recall increases, which is important in settings where flagging human work as AI (false positives) is costly.

4.6 Feature Importance and Analysis

Using the Random Forest component, we compute feature importance scores based on the average decrease in impurity when splitting on each feature. The top five features are:

- Readability score ($\approx 29\%$ importance)
- Average word length ($\approx 27\%$)
- Punctuation density ($\approx 24\%$)
- AI word ratio ($\approx 9\%$)
- Sentence length variation ($\approx 7\%$)

This ranking suggests that structural and stylistic aspects—how complex the sentences are, how long the words are, and how punctuation is used—carry more predictive power than vocabulary diversity alone. AI-generated essays in our dataset tend to occupy a narrower readability range, use longer and more abstract words, and exhibit more regular punctuation patterns. Human texts show greater variability in these features, including simpler language, irregular commas or fragments, and occasional contractions.

We also compare average feature values for human vs. AI texts. AI texts show higher average word length and burstiness, slightly higher punctuation density, higher AI word ratio, and fewer contractions. Humans display larger sentence length variation and somewhat higher pronoun and contraction rates. These differences align with intuitions from both stylometry and qualitative observations of LLM outputs.

4.7 Case Studies

To complement aggregate metrics, we examine specific examples processed by the detector.

A clearly AI-like paragraph with dense technical language and multiple abstract noun phrases (e.g., “the multifaceted nature of contemporary computational paradigms”) receives an AI probability above 80%. Its feature vector shows long average word length, high readability score, moderate AI word ratio, and relatively low sentence length variation, all consistent with the AI profile.

An informal diary-style text (e.g., “I can’t believe how bad traffic was today. Seriously, it took me an hour to go five miles!”) receives a human probability above 80%. It contains contractions, emotional expressions, an exclamation mark, and a mixture of shorter sentences, leading to features closer to the human cluster.

A borderline case starts with a human-like opening (“Honestly, I think the main issue is communication”) but then shifts into a formal, abstract style (“one must consider broader societal implications”). The model outputs an AI probability close to 0.5, indicating uncertainty. The feature vector reflects this mixed nature: moderate AI word ratio and readability score, but still some informal markers. These examples illustrate that the detector’s behavior is consistent with the feature-level explanations.

5 Discussion

5.1 Strengths

The main strength of our approach is that it offers a good balance between performance, efficiency, and interpretability. Despite using only a small set of hand-crafted features and classical classifiers, the detector reaches over 91% accuracy and high AUC scores on a reasonably large and diverse dataset. The model runs quickly, with average inference time on the order of tens of milliseconds per text on a standard laptop, making it suitable for interactive use in educational or content moderation scenarios.

Another advantage is transparency. Because the features are explicit and interpretable, we can provide users with concrete explanations (e.g., “this text has very low sentence length variation and a high AI word ratio”), which is harder to achieve with large neural detectors whose internal representations are opaque.

5.2 Limitations

There are several important limitations. First, our dataset is still biased towards English academic and essay-style texts. Performance may degrade on genres such as poetry, code-mixed text, or social media comments, which exhibit very different stylistic patterns. Second, the detector operates at the level of whole paragraphs or essays and may be less reliable for very short sentences, where stylometric features are less stable.

Third, as LLMs become better at mimicking human imperfections, the stylometric gap between human and AI writing is likely to shrink. Models already support style-transfer and can introduce intentional noise, which may make style-based detectors less effective without continuous adaptation.

Finally, our system is not robust to adversarial authors who intentionally manipulate features (e.g., injecting contractions or random punctuation) to evade detection. Addressing such adversarial behavior may require combining stylometric features with other signals, such as provenance information or watermarking techniques [6].

5.3 Future Work

Future directions include expanding the dataset to cover more languages and genres, incorporating more advanced yet still interpretable features (e.g., shallow syntactic patterns), and studying robustness under paraphrasing and manual editing. Another promising avenue is hybrid systems that combine lightweight stylometric detection with watermark-based signals or transformer-based detectors, leveraging the strengths of each approach.

6 Conclusion

We have presented an AI text detector that distinguishes human-written texts from AI-generated texts using a compact set of stylometric features and an ensemble of traditional machine learning models. On a balanced dataset of nearly 5,000 texts combining real and synthetic sources, the system achieves around 91% test accuracy, with ROC AUC of 0.968 and PR AUC of 0.971, while remaining computationally lightweight.

Our analysis of feature importance and example predictions shows that structural aspects of writing—readability, average word length, punctuation density, and sentence rhythm—play a central role in separating human and AI texts. These findings echo long-standing results in stylometry and authorship attribution and suggest that style-based cues remain valuable even in the era of large language models.

While there are clear limitations in terms of domain coverage and robustness, we believe that interpretable, stylometry-based detectors provide a useful building block for responsible AI text detection in educational and platform settings. They can serve as transparent tools to support human judgment, rather than as opaque oracles, and they can be combined with other techniques as the landscape of AI-generated content continues to evolve.

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